Mathematical Formalization of the Logistic Growth Nonlinear Mixed Effects Model

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Stage 1: Individual-Level Model

Let y_{ij} denote the observed response for individual $i \in \{1, \ldots, n\}$ at time t_j for $j \in \{1, \ldots, J\}$. The intra-individual model describes the relationship between time and response using a logistic growth curve:

$$y_{ij} = f(t_j, \beta_i) + e_{ij} \tag{1}$$

where $\beta_i = (\beta_{i1}, \beta_{i2}, \beta_{i3})^T$ is the vector of subject-specific latent parameters representing the asymptote, inflection point, and scale, respectively. The nonlinear function is defined as:

$$f(t_j, \beta_i) = \frac{\beta_{i1}}{1 + \exp\left(-\frac{t_j - \beta_{i2}}{\beta_{i3}}\right)} \tag{2}$$

The intra-individual measurement errors are assumed to be independent and identically distributed: $e_{ij} \sim \mathcal{N}(0, \sigma^2)$.

Stage 2: Population-Level Model

The inter-individual model describes the distribution of the latent individual parameters β_i across the population:

$$\beta_i = \mu + b_i \tag{3}$$

where $\mu = (\mu_1, \mu_2, \mu_3)^T$ represents the fixed population mean effects. The random effects b_i capture the biological variation between individuals and are modeled as a multivariate normal distribution:

$$b_i \sim \mathcal{N}(\mathbf{0}, \Sigma) \tag{4}$$

where Σ is a 3×3 symmetric positive-definite covariance matrix.

We define the set of population parameters to be estimated as $\theta = \{\mu, \Sigma, \sigma^2\}$.

Likelihood Formulation

Conditional Likelihood (Observations given Latent Variables)

Given a sample of the latent subject-specific parameters β_i and the residual variance σ^2, the likelihood of the observations $\mathbf{y}_i = (y_{i1}, \ldots, y_{iJ})^T$ for individual i is:

$$p(\mathbf{y}_i \mid \beta_i, \sigma^2) = \prod_{j=1}^{J} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_{ij} - f(t_j, \beta_i))^2}{2\sigma^2}\right) \tag{5}$$

The conditional log-likelihood is therefore:

$$\ln p(\mathbf{y}_i \mid \beta_i, \sigma^2) = -\frac{J}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{j=1}^{J}(y_{ij} - f(t_j, \beta_i))^2 \tag{6}$$

Prior Likelihood (Latent Variables given Population Parameters)

The likelihood of the latent parameters β_i under the population prior is determined by the fixed effects μ and covariance Σ:

$$p(\beta_i \mid \mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^3|\Sigma|}}\exp\left(-\frac{1}{2}(\beta_i - \mu)^T\Sigma^{-1}(\beta_i - \mu)\right) \tag{7}$$

The prior log-likelihood is:

$$\ln p(\beta_i \mid \mu, \Sigma) = -\frac{3}{2}\ln(2\pi) - \frac{1}{2}\ln|\Sigma| - \frac{1}{2}(\beta_i - \mu)^T\Sigma^{-1}(\beta_i - \mu) \tag{8}$$

Complete Joint Likelihood

The complete data joint log-likelihood for individual i, encompassing both the observations and the latent parameters, is the sum of the conditional and prior log-likelihoods:

$$\ln p(\mathbf{y}_i, \beta_i \mid \theta) = \ln p(\mathbf{y}_i \mid \beta_i, \sigma^2) + \ln p(\beta_i \mid \mu, \Sigma) \tag{9}$$

Loss Function and Minimization

The objective of the optimization is to minimize the total negative joint log-likelihood over all n individuals. Thus, the loss function is defined as:

$$\mathcal{L}(\theta, \boldsymbol{\beta}) = -\sum_{i=1}^{n}\ln p(\mathbf{y}_i, \beta_i \mid \theta) \tag{10}$$

Because the individual parameters β_i are unobserved latent variables, we utilize a stochastic optimization algorithm combining Markov Chain Monte Carlo (MCMC) sampling with gradient descent. The pseudo-algorithm proceeds as follows:

1. **Initialization**: Initialize the parameters $\theta^{(0)}$ and the latent variables $\beta_i^{(0)}$ for all individuals $i \in \{1, \ldots, n\}$.
2. **Iteration**: For each step $k = 1, 2, \ldots$:
    - **Sampling Step (Metropolis-within-Gibbs)**: For each individual i and each parameter dimension $d \in \{1, 2, 3\}$, propose a new state β_{id}^* from a normal proposal distribution with an adaptive step size. Accept the proposed variable with probability:

$$\alpha = \min\left(1, \exp\left(\ln p(\mathbf{y}_i, \beta_i^* \mid \theta^{(k-1)}) - \ln p(\mathbf{y}_i, \beta_i^{(k-1)} \mid \theta^{(k-1)})\right)\right) \tag{11}$$

    - **Gradient Computation**: Evaluate the gradient of the loss function with respect to the global parameters using the newly sampled latent variables:

$$g^{(k)} = \nabla_\theta \mathcal{L}(\theta^{(k-1)}, \boldsymbol{\beta}^{(k)}) \tag{12}$$

- **Parameter Update**: Update the parameters θ using an adaptive moment estimation algorithm (similar to Adam). Apply an exponentially decaying step size (cooling schedule) γ_k:

$$\theta^{(k)} = \theta^{(k-1)} - \gamma_k \cdot \text{AdamUpdate}(g^{(k)}) \tag{13}$$

3. **Termination**: The algorithm terminates when a predefined convergence criterion is met, such as when the smoothed moving average of the gradients multiplied by the step size falls below a threshold ϵ.